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Research Article

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Concentrations of per- and polyfluoroalkyl substances in vegetables from Sydney and Newcastle, Australia

Siyuan Liu,[a] Luchun Duan,[a,b*] Feng Shi,[a] Gabriel M Filippelli[c] and Ravi Naidu[a,b*]

Abstract

BACKGROUND: This study investigated per- and polyfluoroalkyl substances (PFASs) in 53 fruit and vegetable samples collected from a local wholesale and retail market in Sydney and a local supermarket in Newcastle. As there is limited information about PFAS levels in vegetables on the market, this study aimed to fill this gap and assess potential risks for humans through consumption of these vegetables.

METHODS: QuEChERS extraction – a solid-phase extraction method, a portmanteau word formed from 'quick, easy, cheap, effective, rugged and safe' – followed by enhanced matrix removal–lipid cleaning and liquid chromatography–tandem mass spectrometry analysis were used to detect 30 PFASs in vegetables.

RESULTS: PFOA was detected in 7 out of the 53 samples, with concentrations of 0.038–1.996 ng g^{-1} fresh weight; PFOS was detected in 2 samples only, with concentrations ranging from 0.132 to 0.911 ng g^{-1} fresh weight. PFHxS was not detected in any sample in this study. PFOA and PFOS concentrations measured in vegetables in this study constituted daily intake of 2.03 ng kg^{-1} body weight (BW) and 1.98 ng kg^{-1} BW, respectively, according to recommended daily vegetable intake and BW data from the Australian Bureau of Statistics. The most sensitive population group is girls of 4–8 years of age. These estimated exposure levels represent up to 1.3% of the tolerable daily intake for PFOA (160 ng kg^{-1} BW) and 9.9% for PFOS (20 ng kg^{-1} BW) according to Food Standards Australia New Zealand. Consumption of the vegetables from the study locations poses a marginal risk to human health.

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Keywords: PFAS; vegetable; market survey; risk assessment; TDI

INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) represent a large group of chemicals widely used in industrial and agricultural production. There are huge demands for these chemicals as they are used in pesticide synthesis, textiles, electroplating, personal care products, fire protection, leather, polymers (especially food packaging) and many other fields.[1,2] For simplification, perfluoroalkyl acids including perfluorocarboxylic acids (PFCAs) and perfluorosulfonic acids (PFSAs) are classed into 'short chain' (PFCAs with C4–7 and PFSAs with C4–5) and 'long chain' (PFCAs with C8+ and PFSAs with C6+) in this study, following the definition adopted by the global regulatory and scientific communities.[3-5]

Exposure to PFASs occurs via various pathways, including food and water ingestion, indoor air inhalation and water and soil dermal contact.[6,7] In these exposure pathways, food and water intakes are primary routes of exposure to these compounds.[8] Various studies showed that PFAS contamination in food is present worldwide. Median values of PFOSs in fish samples from a lake in Germany ranged from 34 (in whitefish) to 96 ng g^{-1} (in perch).[9] Other studies conducted in the USA,[7] Spain,[10,11] France,[12,13] Italy,[14] Greece,[15] Czech Republic,[16] Faroe Islands[17]

* *Correspondence to: L Duan or R Naidu, Cooperative Research Centre for Contamination Assessment and Remediation of the Environment (CRC CARE), University of Newcastle, NSW, Australia, E-mail: luchun.duan@newcastle.edu.au (Duan); ravi.naidu@newcastle.edu.au (Naidu)*

a *Global Centre for Environmental Remediation (GCER), University of Newcastle, Callaghan, New South Wales, Australia*

b *Cooperative Research Centre for Contamination Assessment and Remediation of the Environment (CRC CARE), University of Newcastle, Callaghan, New South Wales, Australia*

c *Department of Earth Science, Indiana University–Purdue University, Indianapolis, Indiana, USA*

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and China[18] have reported that PFASs are present in animal products.

As for other foods, a survey[19] of food purchased in Netherlands retail chains showed that crustaceans (0.58 ng g^{-1}) and lean fish (0.31 ng g^{-1}) exhibited high PFOS levels, and vegetables/fruits and flour were important contributors of PFOA intake, whereas milk, beef and lean fish were important contributors of PFOS intake.[20] A study[21] in China's vegetable production base reported that PFASs were found in all investigated vegetables (1.67–33.5 ng g^{-1} dry weight in the edible parts), and that PFBA was the most common contaminant.

Food packages containing PFASs might be one of the sources that lead to risk. In Spain, unpackaged lettuce contained a mean concentration of PFOS at 0.010 ng g^{-1}, whereas for packaged lettuce the value was 0.034 ng g^{-1}.[22] A study in China found that certain milk packaging greatly increased the total PFASs (from 0.178 to 0.6 ng mL^{-1} on average) in dairy products, with PFOA being one of the contaminants transferred from the packaging.[23] However, most studies have focused on paper plates, bowls, boxes, popcorn packaging and fast food containers[24,25] and vegetable packaging with printing has not been studied.

Thus far, most market surveys have focused on meat and seafood and surveys on the presence of PFASs in vegetables are relatively rare. The study reported here aimed to investigate potential exposure to PFASs from consumption of selected vegetables sold by market suppliers in Sydney (from the Sydney Market, the largest wholesale fruit, vegetable and flower market in Australia) and Newcastle (from a local chain supermarket), Australia, so that the associated risks could be estimated. Some package analysis was also conducted to check PFAS migration to vegetables. The QuEChERS (quick, easy, cheap, effective, rugged and safe) method was chosen as the extraction method for this study given the time-efficiency and accuracy it provides.

EXPERIMENT DESIGN

In this study, we sampled 53 vegetable samples (37 types of vegetables) from the Sydney Market and a local supermarket in Newcastle. Sampling information is presented in Table S1. Concentrations of 30 PFASs (detailed in Tables S2 and S3) were measured in this study. An Agilent 1260 Infinity II liquid chromatography (LC) system coupled with an Agilent 6470 triple quadrupole mass spectrometer (MS) with an Agilent Jet Stream thermal gradient focusing electrospray ionization source was the main analytical instrumentation (Tables S4–S6).

Extraction methods

The extraction method consisted of ultrasonication extraction, dispersive solid-phase extraction clean-up and HPLC–MS/MS instrument analysis. Calibration of internal standards and matrix matching by stable isotope labelling helped compensate for any recovery loss.

The samples were transferred to polypropylene cans and freeze-dried for 48 h followed by another 24 h until the weight was constant. Samples were then homogenized with a coffee blender and stored at −20 °C before further usage. To clarify, all samples were analyzed as freeze-dried samples and converted to fresh weight as the final results with moisture data based on the database from the US Department of Agriculture (Table S1).

Extraction and cleaning methods of vegetables are shown in Fig. 1. In view of PFAS exposure risks via vegetables, the edible parts of vegetables were selected as the target of this study.

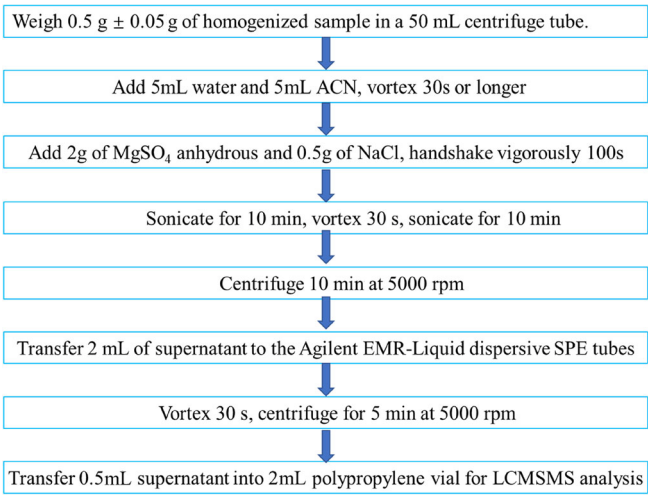


Figure 1. Flowchart of extraction.

For the analysis of PFAS contamination of packaging, the extraction procedure was modified from the vegetable extraction method. In brief, a quarter of a whole package was cut to pieces and transferred into a 50 mL polypropylene centrifuge tube. Around 45 mL of LC–MS grade acetonitrile, which had been confirmed as PFAS-free, was added to these samples to completely submerge the package. The tubes were sonicated for 20 min at 30 °C, vortexed for 30 s and then sonicated again. After centrifuging at 5000 rpm for 5 min, 10 mL of supernatant was transferred to a 15 mL polypropylene centrifuge tube followed by nitrogen blow drying to 1 mL. Concentrated extracts were centrifuged at 8000 rpm for 10 min. An amount of 0.5 mL of supernatant was transferred to a LC vial followed by adding internal standard (5 ng g^{-1}) for LC–MS/MS analysis. The total PFAS value of each bag was calculated according to Eqn (1):

$$\text{Total PFAS per bag} = \text{Instrument concentration} \times \frac{1\,\text{mL}}{10\,\text{mL}} \times 4 \quad (1)$$

LC–MS/MS analytical method

A fast and reliable analytical method using LC–MS/MS was developed and validated for quantitative analysis of 30 PFASs including C4–14 PFCAs, C4–C10 PFSAs, 4–10:2 FTS, 2 FOSAs, 2 FOSEs and 3 FOSAAs. The analysis was performed with a LC–MS/MS system from Agilent.

Fluoropolymers are widely applied in the manufacture of instruments and can result in high PFAS background conditions by leaching of short-chain PFASs (e.g. PFBA from the autosampler or degassing unit during LC–MS/MS analysis). A delay column (50 mm × 4.6 mm, 4 µm; Agilent Infinity Poroshell 120 C18 column) was installed between the mobile phase mixer and the autosampler to trap any background PFASs leaching from the instrument. In addition, traditional LC vials capped with polytetrafluoroethylene/silicone septa may contribute to the evaluated PFAS background condition. To avoid potential PFAS adherence to glass vials, polypropylene vials and snap-caps were used in this study. Detailed instrumental conditions are presented in Tables S4–S6.

Quality assurance

Quality assurance was conducted by regular analyses of quality control samples including reagent blanks, laboratory control

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samples and duplicate samples within each batch. Relative standard deviation (RSD) is the population standard deviation divided by the population mean and was used for calculation of average values among groups. Duplicate assays were performed at a minimum rate of one per batch or 10% whichever was the greater, and relative percent difference (RPD) regarding duplicate analysis was calculated according to Eqn (2):

$$\mathrm{RPD} = \frac{(x_H - x_L)}{\bar{x}} \times 100 \quad (2)$$

where x_H is the highest result, x_L is the lowest result and $\bar{x}$ is the mean result.

Statements of significant differences were based on 95% confidence intervals. RPD values were determined for precision. A continuing calibration verification standard at the mid-calibration range concentration was analyzed for every 20 samples or fewer as the case may be.

The linearity of the instrumental response was evaluated with standard calibration curves with PFAS concentrations ranging from 0.025 to 10 ng g^{-1}. Excellent linearity ($R^2 > 0.995$) was obtained for all target PFAS analytes, where at least 4 points must be used to establish the linear calibration curve or at least 6 points must be fitted for the quadratic calibration curve (Table S7).

RESULTS AND DISCUSSION

Spike recovery study

Spike recovery of PFASs was investigated using two methods. The first method is matrix spike recovery, which involved native PFASs only, and seven clean vegetable samples (carrot, cucumber, lettuce, spinach, potato, tomato and avocado) from the unrelated market that tested not containing those PFAS were utilized to verify the efficiency of the extraction method. Native PFASs were added into lyophilized vegetable sample in a 50 mL centrifuge tube before the extraction procedure. For the native PFAS fortification experiment, internal standard at 5 ng g^{-1} was added before LC–MS/MS analysis. Each sample extraction was performed in triplicate. The matrix spike recovery was calculated to determine the extraction method performance according to Eqn (3):

$$\text{Matrix spike recovery} = \frac{\text{Actual concentrations in spiked sample}}{\text{Theoretical sample concentrations}} \times 100\% \quad (3)$$

Recovery of the eight PFASs is presented in Table 1. Recoveries by contaminants ranged from 71.0% to 128.1% and from 99.7% to 107.1% by matrix, confirming the suitability of this extraction procedure for PFAS analysis in plant samples.

The second method is isotopic PFAS standard spike experiment. To investigate matrixes with matrix effect on PFAS extraction and analysis, 17 vegetable samples (bok choy, cabbage, broccoli, fennel, garlic, garlic bolt, ginger, beetroot, carrot, purple sweet potato, radish, bitter gourd, cucumber, eggplant, green chili, okra and tomato) were included.

For the isotopic PFAS standard spike experiment, direct comparison of the isotopic PFAS standard peak area was made with the peak area of the standard concentration at the same concentration level (5 ng g^{-1}) to calibrate the recovery. The isotopic internal standard spike recovery (ISR) was calculated to determine the extraction method performance according to Eqn (4):

$$\mathrm{ISR} = \frac{\text{Isotope labelled PFAS peak area in vegetable} \times 10}{\text{Isotope labelled PFAS mean peak area in calibration curve}} \times 100\% \quad (4)$$

Recoveries of the 17 vegetable samples are presented in Table 2. Recoveries by matrix ranged from 83.9% to 109.8%, confirming the matrix effect is not obvious under the extraction and cleaning procedure.

PFASs in vegetable samples

The detected PFCA and PFSA concentrations in all 53 vegetables are summarized in Table 3. PFAS constituents from the Sydney market samples and Newcastle supermarket samples are respectively shown in Figs 2 and 3.

PFCAs were frequently detected in 41.5% samples (22 out of 53 samples) (Table 3), with total concentrations up to 1.966 ng g^{-1} detected for rosemary (*Salvia rosmarinus*).

It is evident that short-chain PFCAs were the main contributors to PFCA contamination in vegetables, as 72.7% (16 out of 22 samples) of PFCA-contaminated samples contained the short-chain PFCAs. Samples from leaf and fruit groups tended to contain more short-chain PFCAs compared to other groups as 13 out of the 16 short-chain-contaminated samples are in this groups, or 14 out of 16 if one considers bean sprout (*Vigna radiata*) as fruit. This confirms prior findings,[26,27] in which the short chains, especially PFBA, tended to accumulate in high-moisture plant parts such as leaf and fruit with water transportation in plants. For the same reason, a previous study[26] also used the QuEChERS method on broccoli and especially mentioned the phenomenon that flower part tends to accumulate more PFASs. This corresponded to broccoli in our study, which measured PFBA at 0.189 ng g^{-1}.

PFSA contamination was comparatively rare. Only 3 samples, dill (*Anethum graveolens*), garlic (*Allium sativum*) and onion (*Allium cepa*), were contaminated out of all 53 samples. Dill and garlic are contaminated by PFOS at concentrations up to 0.991 ng g^{-1}, which is even higher than the concentration of 0.16 ng g^{-1} for PFOS + PFHxS detected in lettuce samples from several farms near a large fluoride factory in China.[28] This may be because some lipid-related proteins tend to accumulate PFOS. A study[29] has demonstrated that the fatty acid binding proteins in plants may be the main factor that influences PFOA and PFOS distribution and transportation. Protein and lipid contents in dill are 3.46% and 1.12% by mass, and in garlic are 6.36% and 0.50%, higher than those in other vegetables such as lettuce (contains protein at 0.90% and lipid at 0.14%), tomato (0.88% and 0.20%) and radish (0.68% and 0.10%).[30] It was pointed out that PFOA and PFOS tend to distribute in protein-rich tissues, rather than just in lipid-rich tissues, as lipid may compete with PFOA and PFOS for bonding sites at those proteins.[29] The rosemary sample showed the highest PFOA concentration in our study at 1.966 ng g^{-1}, which may be because of its high protein and lipid levels (protein at 3.31% and lipid at 5.86% by mass)[30] or package contamination, which was investigated in a subsequent study.

Although previous studies reported that chain length will influence the PFSA plant intake and distribution just like PFCA,[26,27,31] we could not draw any conclusion on this topic due to low sample detection number.

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Table 1. Recovery of matrix spiked surrogate

Matrix	PFAS recovery ± RSD (%)							Average recovery (%)
	PFBA	PFHxA	PFOA	PFBS	PFHxS	PFOS	FOSA	
Avocado	99.2 ± 2.4	99.1 ± 1.3	116.4 ± 4.5	101.1 ± 1.0	106.9 ± 4.1	98.0 ± 5.1	114.6 ± 1.2	105.0 ± 7.4
Lettuce	99.0 ± 7.1	99.2 ± 5.5	117.8 ± 3.8	99.6 ± 3.7	104.4 ± 5.0	106.9 ± 11.1	109.5 ± 7.8	105.2 ± 6.6
Cucumber	83.1 ± 1.4	97.1 ± 0.0.7	128.1 ± 3.4	103.3 ± 3.4	103.1 ± 8.6	71.0 ± 0.7	112.3 ± 2.1	99.7 ± 18.7
Lettuce	98.4 ± 4.5	110.2 ± 5.0	109.7 ± 6.9	103.9 ± 1.4	103.7 ± 17.4	116.0 ± 3.6	107.5 ± 5.4	107.1 ± 5.3
Spinach	79.5 ± 2.6	107.7 ± 3.2	121.2 ± 17.4	101.6 ± 2.9	102.0 ± 5.1	110.8 ± 12.1	100.4 ± 6.6	103.3 ± 12.3
Potato	95.1 ± 6.3	97.0 ± 2.3	114.7 ± 1.1	98.9 ± 2.4	101.7 ± 3.7	102.5 ± 1.9	99.2 ± 2.9	101.3 ± 6.3
Tomato	105.6 ± 2.6	99.4 ± 3.7	114.3 ± 7.0	96.7 ± 5.6	97.9 ± 7.4	97.0 ± 8.4	101.2 ± 5.8	101.7 ± 6.2
Average recovery (%)	94.3 ± 10.0	101.4 ± 5.2	117.5 ± 5.0	100.7 ± 2.5	102.8 ± 2.7	100.3 ± 14.6	106.4 ± 5.8	—

Table 2. Recovery of isotope as surrogate

Vegetable group	Matrix	Spike recovery (±RSD%)						
		PFBA-$^{13}C_4$	PFOA-$^{13}C_4$	PFBS-$^{13}C_3$	PFHxS-$^{18}O_2$	PFOS-$^{13}C_4$	6:2 FTS-$^{13}C_2$	FOSA-$^{13}C_8$
Leaf	Bok choy	44.8	97.0	95.9	73.7	93.3	109.8	76.3
	Cabbage	108.7	81.2	83.4	84.1	101.7	53.0	100.8
Flower	Broccoli	78.6	104.2	102.1	106.2	94.4	119.0	94.3
Stem	Fennel	135.1	101.9	88.4	108.0	65.8	103.1	93.8
	Garlic	104.2	95.9	94.2	88.6	50.8	71.9	92.9
	Bolt	119.0	94.7	98.3	83.4	71.6	105.2	81.9
	Ginger	126.6	127.4	114.2	94.3	114.0	118.8	242.5
Root	Beetroot	108.8	80.5	85.8	81.5	78.5	86.8	73.5
	Carrot	46.5	144.0	110.3	79.9	79.0	161.5	85.9
	Purple sweet potato	79.9	78.1	110.7	99.4	93.7	83.1	99.9
	Radish	88.6	116.5	89.8	54.8	105.8	112.9	68.9
Fruit	Bitter gourd	117.2	191.6	103.4	77.0	81.9	266.5	103.6
	Cucumber	137.7	147.0	125.0	119.8	79.1	152.5	105.4
	Eggplant	147.3	87.2	79.4	98.9	87.2	99.8	90.2
	Green chilli	134.0	113.3	110.6	112.2	67.3	94.6	98.2
	Okra	130.4	94.1	112.4	84.5	86.7	55.7	72.0
	Tomato	132.7	111.3	98.5	88.9	75.3	71.0	113.1
Average recovery		108.2 ± 28.6	109.8 ± 26.7	100.1 ± 12.5	90.3 ± 17.8	83.9 ± 18.9	109.7 ± 45.5	99.6 ± 39.1

Table 3. PFCAs and PFSAs detected in both Markets

Group	Total PFCA concentration (ng g^{-1} fresh weight)	PFCA contaminated sample (Group sample number)	Short-chain PFCA contaminated sample (PFCA contamination sample)	Total PFSA concentration (ng g^{-1} fresh weight)	PFSA contaminated sample number (Group sample number)	Short-chain PFSA contaminated sample (PFSA contamination sample)
Leaf	<LOQ ~ 1.966	62.5% (10/16)	80% (8/10)	<LOQ ~ 0.132	6.3% (1/16)	0.0% (0/1)
Flower	0.189	100% (1/1)	100% (1/1)	<LOQ	0.0% (0/1)	N/A
Stem	<LOQ ~ 0.217	27.3% (3/11)	66.7% (2/3)	<LOQ ~ 0.991	18.2% (2/11)	50.0% (1/2)
Root	<LOQ ~ 0.270	25.0% (2/8)	0.0% (0/2)	<LOQ	0.0% (0/8)	N/A
Fruit	<LOQ ~ 1.008	35.3% (6/17)	83.3% (5/6)	<LOQ	0.0% (0/17)	N/A
Total	Up to 1.966	41.5% (22/53)	72.7% (16/22)	Up to 0.991	5.7% (3/53)	33.3% (1/3)

N/A, not applicable.

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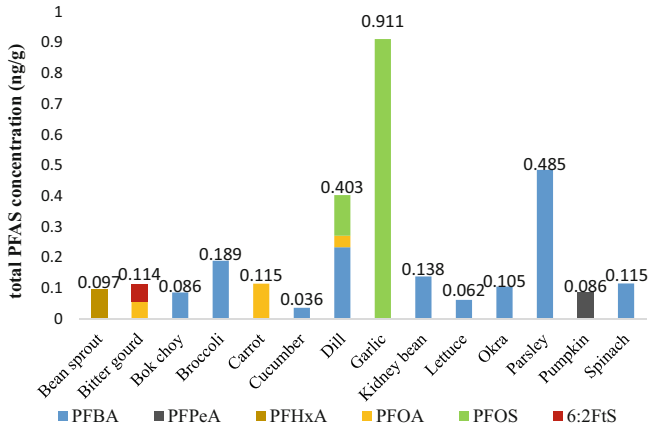


Figure 2. PFASs detected in vegetable samples from the Sydney Market (ng g^{-1} fresh weight).

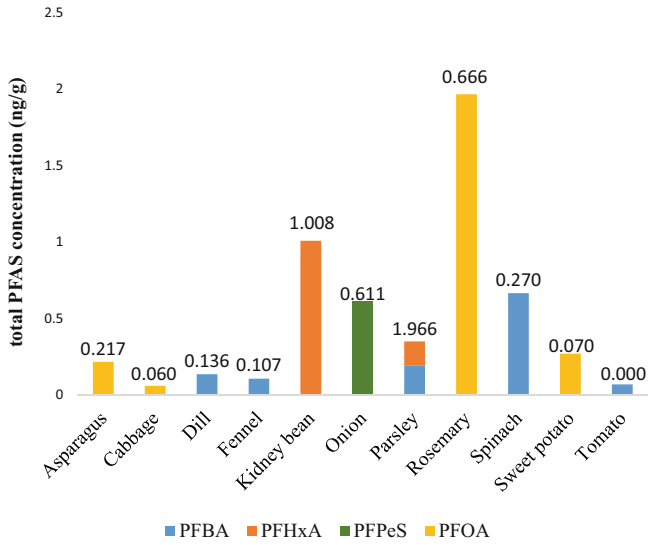


Figure 3. PFASs detected in vegetable samples from Newcastle supermarket (ng g^{-1} fresh weight).

Figure 2 shows all the samples with detected PFASs from the Sydney Market. In Fig. 2, PFBA is the main contaminant among all the samples, while PFPeA and PFHxA were detected in only one vegetable sample (pumpkin and bean sprout), respectively. PFOS contamination in garlic and dill is obvious while other contamination contributions are made by PFCAs. Bitter gourd contained PFOA and 6:2 FTS contamination. However, given the ambiguity regarding the matrix effect (maybe alkaloid influence) there may be some uncertainty with these concentrations in bitter gourd. For this reason, bitter gourd data were not considered in the section on daily intake calculations.

Figure 3 shows the PFOA and some short-chain PFAS distribution in the local supermarket samples. PFOA was the only long-chain PFAS detected in the local supermarket vegetables. Kidney bean (*Phaseolus vulgaris*), onion (*Allium cepa*) and spinach (*Spinacia oleracea*) showed the highest concentrations of PFHxA (1.008 ng g^{-1}), PFPeS (0.611 ng g^{-1}) and PFBA (0.666 ng g^{-1}), respectively. This is significantly different from the PFAS concentrations measured in samples from the Sydney Market where PFOS was detected in two samples, PFOA was rare and PFBA was the main contributor of PFAS contamination. We believe that further research on this difference is needed.

There are many potential sources of PFASs that can contaminate vegetables. For example, planting fields with biosolid as soil amendments could lead to PFAS contamination,[32-34] irrigation water tends to contain short-chain PFAS with higher mobility compared to long-chain ones,[27,35,36] some volatile or particle-bound precursors could be metabolized into PFAS contaminants once being absorbed by plant leaves via the atmosphere[37] or pollution during transportation. However, such information could not be collected in this study and only the packaging as a potential source of contamination was considered. So, the testing of the Newcastle supermarket commercial packages is considered as the Sydney Market samples do not have the commercial packaging from the stall owner.

In our study, detected PFAS concentrations were slightly higher than those reported in the European Union (EU) project PERFOOD[38] in which PFOA concentrations in vegetables from four European countries ranged from 0.002 to 0.025 ng g^{-1} on a fresh weight basis. In that study, PFOA was the most common PFAS constituent in vegetables from Norway (80%), Italy (47%), Belgium (50%) and Czech Republic (7%). Another study from China[39] reported similar trends with PFOA being the most common PFAS contaminant detected in all 26 vegetables, ranging from 0.023 to 0.153 ng g^{-1}. From their study, this level of PFAS contamination seemed not to be harmful to human health, but calculation according to Australian standard is still needed.

From the results we can see the short-chain PFASs are more commonly detected than long-chain PFASs in vegetables, and PFSAs are less common in plant samples than PFCAs. This is consistent with the results from former studies.[31,35,40] Another finding is that the most common contaminant was PFBA (detected in 14 out of 53 samples) in all vegetable samples. FOSA, FOSAA and FOSE chemicals were not detected in the vegetable samples.

Potential PFAS contamination from packaging

Compared with samples collected from the Sydney Market, samples collected from the local supermarket in Newcastle were detected more frequently with PFOA, with concentrations as high as 1.966 ng g^{-1}. These vegetables, unlike those from the Sydney Market, were packaged in plastic bags (Fig. S1). It is well known that PFASs are present in a variety of food packaging.[24,41,42] Therefore, subsequent examination of PFASs in these food packages was performed as a suspected source of PFAS contamination of these vegetables. Details of the extraction method for food packages are presented in the section on Extraction methods. Generally speaking, large packages were for leaf vegetables, medium for fruit vegetables and small for herbs and spices.

All food packages were found with detectable PFASs, ranging from 0.892 to 1.038 ng g^{-1} (Table 4). Figure 4 further illustrates that PFOS was the dominant contaminant, contributing to 30–72% of total PFASs followed by PFOA and PFBA, contributing to 6–24% and 3–20% of the total PFASs, respectively. PFHxS was only found in the small packages used for herb and spice vegetables; PFDoA, PFTrDA and MeFOSA were only found in the medium and large packages.

It is evident from Table 5 that among all packaged vegetable samples, only four showed bioaccumulations of PFCAs. Regarding possible contamination by the packaging, large packages might have been a source of PFBA, medium packages would have made no additional contribution to PFASs and the small packages might have been a source of PFOA for rosemary. However, for most of

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Table 4. PFAS concentrations in three types of food package in ng g^{-1} (mean) with RSD (%) in parentheses

Package size	PFBA	PFOA	PFHxS	PFOS	Sum of other PFAS[a]	Total PFASs
Large	0.203 (7.3)	0.246 (4.9)	N/A	0.311 (6.2)	0.278	1.038
Medium	0.140 (5.7)	0.057 (3.4)	N/A	0.644 (7.1)	0.052	0.892
Small	0.034 (11.6)	0.235 (1.8)	0.405 (0.9)	0.575 (1.9)	N/A	1.249

[a] Other PFASs include PFDoA, PFTrDA and MeFOSA.

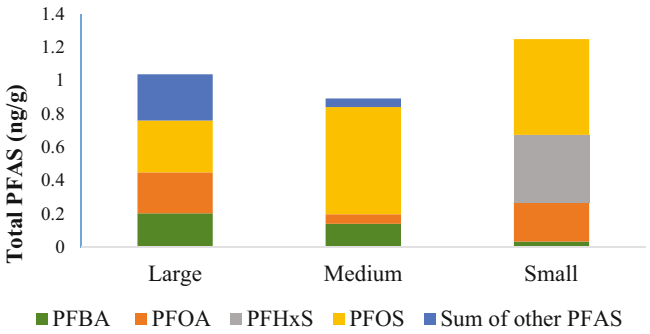


Figure 4. PFASs detected in vegetable package bags.

the vegetables in our study, there was no obvious relationship between the package type and vegetable contamination. An earlier study[43] suggested that food packaging bags had fluorotelomer coating/additive containing PFOA primarily on the outside of the bags. As the migration of PFASs from food-contact surfaces to food increases with increasing temperature, prolonged exposure time and the presence of emulsifiers,[44] simple contact may not lead to severe PFAS migration.

Daily intake calculations

In the estimation of the human intake risk based on the PFAS concentration data, uncertainty associated with concentration level was taken into account by using the upper bound approach in the exposure estimation (Table S8), in which any PFAS concentration below the limit of quantitation (LOQ) was represented by the value for LOQ to calculate a very conservative daily intake.[38]

The daily estimated intake of PFAS in terms of ng kg^{-1} body weight (BW) per day was estimated using the respective PFOA or PFOS average concentrations.

We followed a prior daily food consumption survey[45] and divided the vegetable samples into four main subgroups (other vegetables, starchy vegetables, green and brassica vegetables, and orange vegetables) to obtain the average amounts, in ng g^{-1}, of vegetable PFAS intake. Equation (5) represents the mean PFAS intake value of one selected consumption subgroup, based on all real samples from both the Sydney Market and Newcastle supermarket:

$$\text{Average PFAS concentration} = \frac{\Sigma(\text{PFAS concentratoin in each vegetable sample by UB})}{\text{number of samples}} \quad (5)$$

The minimum recommended number of servings of vegetables for Australian people and the size of one serving (one serving of vegetables is 75 g) were obtained from Australian dietary guidelines. Then, the recommended vegetable daily intake amount (in grams) for different age groups can be estimated (details presented in Table S9).

BW values (in kg) by age and gender for average Australian people were obtained from the 2017 to 2018 national survey (sub-tables 8 and 16),[46] and are listed in Table S9.

Daily PFAS intake was calculated using Eqn (6):

$$\text{Daily PFAS intake} = \frac{\Sigma(\text{Average PFAS conc.} \times \text{GW}) \times \text{DI}}{\text{BW}} \quad (6)$$

where: Average PFAS conc. is the result of Eqn (5). GW is the weight of subgroup, one of the following four vegetable subgroups:

- Other vegetables (major contributors in this category are tomato/tomato products, onions, cucumbers, zucchini and mushrooms) (41.2%)
- Starchy vegetables (mostly from potatoes and potato products) (27.6%)
- Green and brassica vegetables (broccoli, lettuce, spinach, etc.) (15.8%)
- Orange vegetables (carrot, pumpkin, etc.) (10.0%).

DI is the daily intake amount, including recommended value and actual average value, from Australian dietary guidelines, where one serving of vegetables equals 75 g.[47] BW is average body weight, from 4364.0.55.001 – National Health Survey: First Results, 2017–2018.[46]

Food Standards Australia New Zealand (FSANZ) has established a tolerable daily intake (TDI) of 20 ng kg^{-1} BW for PFOS and 160 ng kg^{-1} BW for PFOA. There was insufficient toxicological and epidemiological information to justify establishing a TDI for PFHxS.[48] TDI was compared with the calculated daily intake to illustrate the PFAS exposure risk by vegetable intake.

PFAS exposure was calculated using daily recommended servings (one serving of vegetables is 75 g) by age and sex group, as reported in Table 6 and using the calculation provided above. Risk was estimated by comparing the estimated daily intake with the TDI of 20 ng kg^{-1} BW for PFOS and 160 ng kg^{-1} BW for PFOA from FSANZ.[48] It is evident from Table 6 that for young children, PFOA and PFOS consumption from vegetables (1.97–2.03 and 1.93–1.98 ng kg^{-1} BW day^{-1} respectively for boys and girls of 4–8 years) represented 1.2–1.3% and 9.7–9.9% of the respective TDI values, suggesting no obvious PFAS exposure risk by vegetable intake.

From 4364.0.55.001 – National Health Survey: First Results, 2017–2018 (sub-tables 12 and 17),[46] it is evident that most Australians do not meet recommended amounts of vegetable intake and actual vegetable daily intake is far below the recommended average of five servings a day, at just 2.3 serves on average, which is about half the recommended value.[45] Therefore, the

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Table 5. Detected PFASs in vegetable samples against package types

Package size	Matrix	PFBA (ng g^{-1})	PFHxA (ng g^{-1})	PFOA (ng g^{-1})	Total PFCAs (ng g^{-1})
Large	Brussels sprouts	—	—	—	—
	Celery sticks	—	—	—	—
	Lettuce	—	—	—	—
	Parsley	0.194	0.156	—	0.350
	Spinach	0.666	—	—	0.666
Medium	Avocado	—	—	—	—
	Green chilli	—	—	—	—
Small	Bird eye chilli	—	—	—	—
	Curry leaf	—	—	—	—
	Dill	0.136	—	—	0.136
	Lime leaf	—	—	—	—
	Rosemary	—	—	1.966	1.966
	Thai basil	—	—	—	—

Table 6. Total PFAS daily intake based on examined vegetable samples and recommended vegetable daily intake amounts

Gender	Subgroup	Age group (years) 2–3	4–8	14–18	19–50	51–70	70+
Male	Total PFOA (ng day^{-1})	28.1	50.5	61.8	67.4	61.8	56.2
	Total PFOS (ng day^{-1})	27.4	49.3	60.3	65.8	60.3	54.8
	Minimum average BW (kg)[a]	17.0	25.6	56.3	81.1	90.4	73.9
	PFOA daily intake (ng kg^{-1} BW day^{-1})	1.65	1.97	1.10	0.83	0.68	0.76
	Contribution of TDI value (%)	1.0	1.2	0.7	0.5	0.4	0.5
	PFOS daily intake (ng kg^{-1} BW day^{-1})	1.61	1.93	1.07	0.81	0.67	0.74
	Contribution of TDI value (%)	8.1	9.7	5.4	4.1	3.4	3.7
Female	Total PFOA (ng day^{-1})	28.1	50.5	56.2	56.2	56.2	56.2
	Total PFOS (ng day^{-1})	27.4	49.3	54.8	54.8	54.8	54.8
	Minimum average BW (kg)[a]	16.3	24.9	54.2	65.9	75.1	62.0
	PFOA daily intake (ng kg^{-1} BW day^{-1})	1.72	2.03	1.04	0.85	0.75	0.90
	Contribution of TDI value (%)	1.1	1.3	0.7	0.5	0.5	0.6
	PFOS daily intake (ng kg^{-1} BW day^{-1})	1.68	1.98	1.01	0.83	0.73	0.89
	Contribution of TDI value (%)	8.4	9.9	5.1	4.2	3.7	4.5

[a] Minimum average BW is considered the lowest average BW of the related age group in Table S9. For example, weight of age 2–3 in this table is considered age 2–4 in Table S9, weight of age 19–50 in this table is considered age 18–24 in Table S9. This was intended to estimate the maximum risk.

PFAS exposure risk is lower than the calculated value in Table 6, indicating minimum contribution of vegetable PFAS to human PFAS exposure.

CONCLUSION

This study included 53 samples from the Sydney Market and a local chain supermarket in Newcastle, with 30 PFASs detected with the QuEChERS method and LC–MS/MS analysis. PFCAs were detected in 41.5% of samples (22 out of 53) with concentrations ranging from 0.036 to 1.966 ng g^{-1}. PFSAs were relatively rare and detected in only 3 samples (5.6%) with concentrations ranging from 0.132 to 0.911 ng g^{-1}. Package analysis revealed no evidence of migration of PFASs from packaging into vegetables.

From exposure calculation results, it is apparent that there is limited or no risk of PFOA and PFOS intake under recommended or actual intake values of the Australian National Bureau of Statistics under the guidance of TDI from FSANZ in 2017, which are 20 ng kg^{-1} BW day^{-1} for PFOS and 160 ng kg^{-1} BW day^{-1} for PFOA from vegetable consumption.

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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